

Parsa Abbasian

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EDUCATION

York University

Toronto, ON

Bachelor of Engineering (B.Eng.) in Computer Engineering

Sep. 2025 – Apr. 2029

- **Academic Standing:** GPA: A (Top-tier academic performance)
- Expected specialization in Artificial Intelligence & Machine Learning.
- Relevant Coursework: Applied Calculus I & II, Discrete Mathematics, Linear Algebra, Programming for Engineers.

EXPERIENCE

Undergraduate Researcher — Space Geodesy Lab

Jan. 2026 – Present

York University — Research Fellowship

Toronto, ON

- Conducting data-driven research on extreme weather events (Hurricanes) using satellite observations and climate model datasets.
- Engineering scientific programming workflows in Python for automated data acquisition and multi-dimensional analysis.
- Developing AI-driven machine learning models to characterize hurricane evolution and predict infrastructure impacts.
- Working under the supervision of Dr. Spiros Pagiatakis on environmental risk modeling and storm prediction reliability.

Vice President — MBH&Co.

Jan. 2026 – Present

B2B Subscription-Based Digital Business Directory Platform

Toronto, ON

- Co-founded and architected a B2B subscription-based digital directory platform enabling businesses to register, manage profiles, and access other verified member businesses through gated access control.
- Engineered dynamic member dashboards using Wix Velo (JavaScript) to manage subscription tiers, listing visibility, and role-based permissions via REST APIs.
- Designed and implemented a scalable data ingestion pipeline integrating automated CSV workflows with structured CMS collections to streamline business onboarding.
- Developed subscription-gated access and validation systems with integrated analytics tracking to monitor engagement, subscription activity, and vendor interactions.

Class Representative & Peer Note-Sharing Volunteer

Sep. 2025 – Present

Student Accessibility Services (SAS), York University

Toronto, ON

- Represented students in MATH 1013, MATH 1028, and PHYS 1801, acting as a liaison between students and instructors.
- Organized calculus review sessions supporting academic performance and accessibility standards.

Event Coordinator

Sep. 2025 – Dec. 2025

Iranian Students Association at York University (ISAYU)

Toronto, ON

- Planned and coordinated cultural events, managing logistics, university policy compliance, and safety protocols.

PROJECTS

Undercut | Go, FastAPI, Next.js, PostgreSQL, Docker

Jan. 2026 – Present

- Developed a live full-stack platform aggregating thousands of GTA car listings, utilizing a concurrent **Go** scraper to identify underpriced vehicles in real-time.
- Engineered a data-driven **Market Analysis** engine providing a 5-year projected Total Cost of Ownership (TCO) comparison, factoring in depreciation and fuel efficiency.
- Implemented an automated **Deal Grading** system (S-Tier to F-Tier) using quantitative pricing models to benchmark listings against Fair Market Value (FMV).
- Leveraged **Google Gemini 1.5 Flash** for multi-modal analysis of seller descriptions, providing automated red-flag detection and personalized negotiation scripts.
- Designed a responsive frontend with **Next.js 14** and **Tailwind CSS**, featuring interactive market visualizations and a persistent "Garage" for user-saved deals.

HawkEye | *Python, Flask, Google Gemini API, Docker, Tailwind CSS*

Jan. 2026

- Built an AI-powered multi-modal analysis tool that converts raw inventory videos into structured data listings with automated summarization.
- Integrated **Google Gemini 2.5 Flash** to perform advanced visual classification and natural language extraction from video frames.
- Engineered a custom **FFmpeg** pipeline for high-efficiency frame extraction and timestamp analysis to optimize AI processing time.
- Containerized the entire application using **Docker** to ensure environment parity and scalable deployment on Render.

AlgoDrift | *Next.js, TypeScript, Tailwind CSS, Framer Motion*

Dec. 2025

- Developed an interactive web-based platform for visualizing fundamental data structures and algorithms using **Framer Motion** for fluid state transitions.
- Implemented dynamic, real-time simulations for Linear Search, Binary Search, and Bubble Sort, allowing users to control execution speed and data input.
- Designed a responsive UI with step-by-step execution highlighting to improve pedagogical clarity and user engagement.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Go, C++, SQL.

Web Development: React.js (Next.js, Vite), FastAPI, Flask, Node.js, Tailwind CSS, Wix Headless, Supabase.

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, OpenCV, Google Gemini API, Pandas, NumPy.

Tools & Platforms: Docker, Git/GitHub, Vercel, Render, Linux/Bash, PostgreSQL.

Soft Skills: Technical Leadership, Hackathon Collaboration, Research Analysis, Problem-Solving.